Ariane - HumanNUM | Journée de travail du GT 4
Analyses littéraires assistées par ordinateur

ANNOTER AUX CÔTÉS D'UN GRAND MODÈLE DE LANGAGE

leur rôle dans la reconnaissance d'entités nommées dans des romans

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THE PROJECT

- Among the first NER initiatives to focus on the annotation of full French novels.
- The endgame is to create and publish tools to assist the study and modelisation of literary texts.
- Annotators : Ùna Faller, Romaric Parnasse , Perrine Maurel (ObTIC).

OBSERVATOIRE DES TEXTES, DES IDÉES ET DES CORPUS

INTRODUCTION

Large Language Models [LLM] are able to produce textual content fitting fixed criteria, with a state of the art degree of accuracy and realism (OpenAI, 2023).
As such, studies have considered using them for data augmentation and generation (Whitehouse et al., 2023).
We have striven to use LLM in the task of Named Entity Recognition [NER] (Bogdanov et al, 2024).

NAMED ENTITY RECOGNITION

- NER consists in identifying in a text and tagging "named entities", which are objects bearing a proper noun, such as characters or specific locations.
- The consensus of multiple annotators, even through unqualified crowdsourcing, has proven efficient in providing accurate results (Nowak and Rüger, 2010).
- LLM can thus be considered as possible annotators.

OBJECTIVES

- I. Create a Gold Standard for NER in (French) novels
- II. Train an efficient NER model
- III. Evaluate LLM's efficiency in NER tasks
- IV. Use LLM to create a Silver Standard

CORPUS

- Bel-Ami, Maupassant
- Eugénie Grandet, Balzac (de)
- Germinal, Zola
- Le Rouge et le Noir, Stendhal
- Madame Bovary, Flaubert
- Les Trois mousquetaires, Dumas
- Notre-Dame de Paris, Hugo

Due to the abundance of novels in the public domain, we do not need to generate new texts to annotate.

HUMAN PROTOCOL

- **First annotation campaign**

  Reviewing the novels independantly from other annotators with only simple, limited guidelines.

- **Second annotation campaign**

  The new guidelines focused on solving most annotators' disagreement throughout the corpus.

- **Third annotation campaign**

  Adjusting the guidelines one last time to fit the specific needs of the corpus.

- **Final review**

  Reviewing and correcting.

POTENTIAL ISSUES FOR ANNOTATORS AND AI

● Entity delimitations

Should we annotate the largest mention, opt for nested mentions, or focus explicitely on the smallest mention ?
Example : “the guard company of Mr Essart”

● Style differences

Some authors used common noun more than proper nouns to refer to their characters. Guidelines need to be adaptative.
Exemple : “Esmeralda” and “the egyptian”

● Cultural subtext

One specific intertextual example convinced us to annotate the common noun “the recluse” even tho it rarely shows up in the original text.

MODELS

Llama 3 : 8B

- Very inefficient.
- Unresponsive to prompt evolution.

Llama 3 : 70B

- Lukewarm results.
- Adaptative to the different prompts that were submitted.
- Overall promising.

FINAL PROMPT

You must verify the work of a model trained for NER on seven different texts (Bel-Ami, Eugénie Grandet, Germinal, Le Rouge et le Noir, Les Trois Mousquetaires, Madame Bovary et Notre-Dame de Paris) : during this process, think step by step while taking into account the morpho-syntaxic nature of the different elements constituting the named entities. The model tagged the named entities if and if only they were characters (PERS), locations (LOC) or organisations (ORG) : it was not meant to annotate dates, literary works' titles... The model gave you the informations as a json format, which contain seven elements : each elements has for attributes a unique id, the annotated text and the list of the entities and their metadata as follows (Entity, Label, first character's position, last character's position). Check if each entity exists in the text, if it can be classified as a named entity and if the label (PERS, ORG, LOC) that is affiliated is valid. Otherwise, bring the mistakes to my attention. For example : the entity ('Parisians', 'LOC', 83, 92) would not be valid since 'Parisians' isn't a proper noun. Here is the json object sent by the model :

You must verify the work of a model trained for NER on seven different texts (Bel-Ami, Eugénie Grandet, Germinal, Le Rouge et le Noir, Les Trois Mousquetaires, Madame Bovary et Notre-Dame de Paris)

We state the main task to accomplish, to put the model on the right track from the start. Using “you” comes down to asking the model to play a role, which is helpful (Meskó, 2023).

You must verify the work of a model trained for NER on seven different texts (Bel-Ami, Eugénie Grandet, Germinal, Le Rouge et le Noir, Les Trois Mousquetaires, Madame Bovary et Notre-Dame de Paris)

We pretend the annotated data was created through machine learning. The model tends to be more diplomatic and less critical if the data is said to be of human origin.

You must verify the work of a model trained for NER on seven different texts (Bel-Ami, Eugénie Grandet, Germinal, Le Rouge et le Noir, Les Trois Mousquetaires, Madame Bovary et Notre-Dame de Paris)

The model can struggle to identify the origin of the text, so providing this information in the prompt stops it from guessing and allows it to eventually replace the corpus in a cultural context.

during this process, think step by step while taking into account the morpho-syntaxic nature of the different elements constituting the named entities.

Asking the model to “think step by step” improves its performance in the case of math problems (Heston and Khun, 2023).

during this process, think step by step while taking into account the morpho-syntaxic nature of the different elements constituting the named entities.

This line was added to help the model understand that capitalization was not to be taken into account ["L'égyptienne" and "l'égyptienne"].

The model tagged the named entities if and if only they were characters (PERS), locations (LOC) or organisations (ORG) : it was not meant to annotate dates, literary works' titles...

We inform the model of the different categories and the specific tag associated with them.

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We provide general cases of false positive that may appear in the annotation data set, for example leftovers from a previous campaign that have not been updated during corrections.

The model gave you the informations as a json format, which contain seven elements : each elements has for attributes a unique id, the annotated text and the list of the entities and their metadata as follows (Entity, Label, first character's position, last character's position).

Explaining each element and attribute found in the json file.

Check if each entity exists in the text, if it can be classified as a named entity and if the label (PERS, ORG, LOC) that is affiliated is valid.

We specify the exact task to the model, with more details than in the beginning of the prompt.

For example : the entity ('Parisians', 'LOC', 83, 92) would not be valid since 'Parisians' isn't a proper noun.

By providing an example of a possible error, we apply the "contrastive learning" method (Shorten et al., 2021), where the model focuses on negative samples.

Here is the json object sent by the model :

We then provide the full json file and specify its starting point to the model.

ARE LLMS RELEVANT?

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WHAT'S NEXT ?

- **Adressing class imbalance**

  LLMs have been put forward as a possible way to mitigate class imbalance (Suhaeni et al., 2023).

- **Insist on negative data samples**

  Submitting errors made by human annotators to the model could prevent it from repeating those same mistakes.

- **Data generation and reciprocal evaluation**

  After having the model evaluate our work, we aim to exchange the roles and evaluate its annotations.

THANK YOU

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